

Supporting Information

PyAutoFEP: An Automated FEP Workflow for GROMACS Integrating Enhanced Sampling Methods

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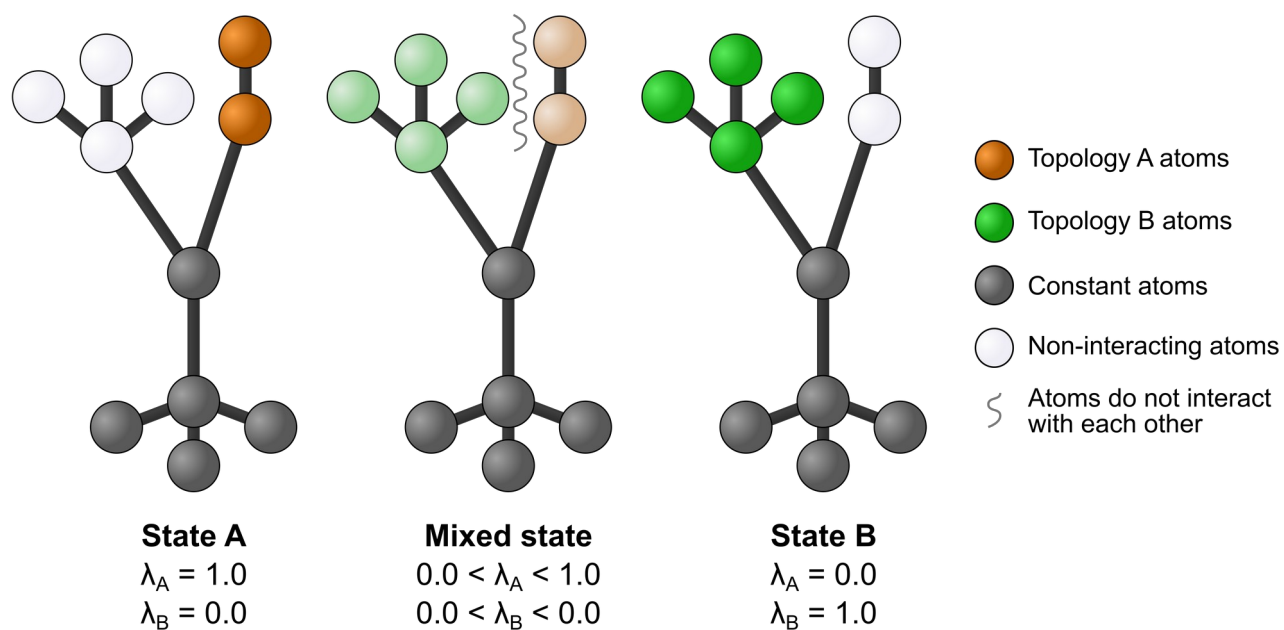


Figure S1 - Representation of the dual topology perturbation model. The atoms belonging only to end state A are represented as non-interacting particles in end state B. In the mixed states, atoms from state A do not interact with atoms from state B, and are partially decoupled from the system (light colors).

Scoring function for the optimal map generator algorithm

The cost of a putative solution during the perturbation optimization is **Tot_{cost}**, given by:

$$\begin{aligned}L_{cost} &= \sum_{node_i} \sum_{node_j} (d_{ij})^{l_e} \\D_{cost} &= e_m \sum_{node_i} (f_{deg}(node_i) - c_d)^{e_p} \\P_{cost} &= p_m \sum_{edge_i} (f_{cost}(edge_i))^{p_e} \\Tot_{cost} &= L_{cost} + D_{cost} + P_{cost}\end{aligned}$$

Where **L_{cost}** is the length term; **d_{ij}** is the distance, in number of edges, between nodes **i** and **j**; **l_e** is length cost exponent, **D_{cost}** is the degree term; **f_{deg}(node_i)** is the degree of node **i** (i.e. the number of edges connected to node **i**), **c_d** is the degree target constant (the ideal number of edges per-node), **e_m** is the degree cost multiplier, **e_p** is the degree cost exponent, **P_{cost}** is the perturbation cost, **p_m** is the perturbation multiplier, **f_{cost}(edge_i)** is the cost of **edge_i** (the number of atoms perturbed during the correspondingly perturbation), and **p_e** is perturbation exponent. The equation was inspired by the Lead Optimization Mapper (LOMAP)¹ and the ant colony optimization works by Dorigo².

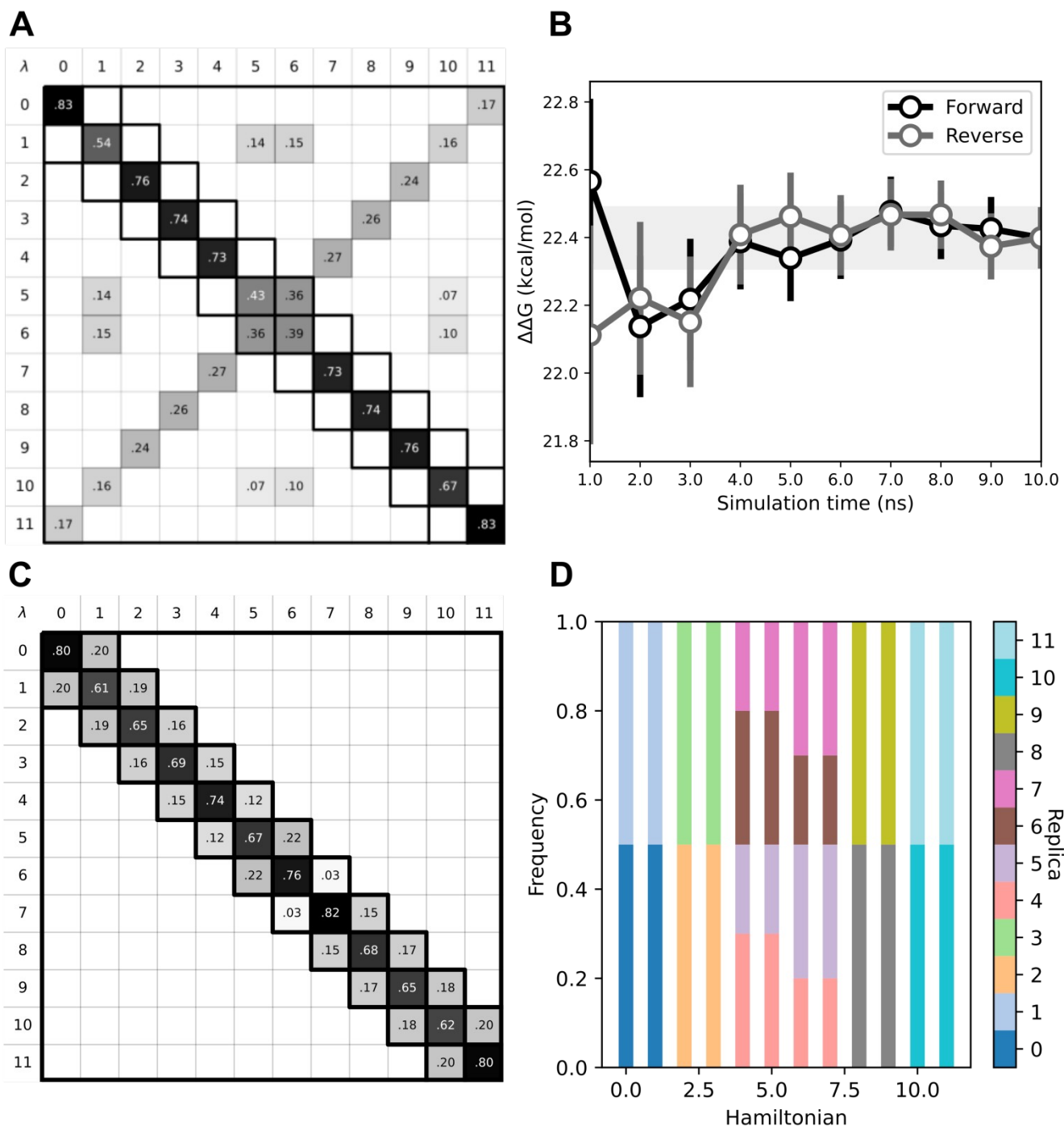


Figure S2 – Example plots generated by PyAutoFEP analysis scripts. (A) Overlap matrix estimated using the pymbar implementation of MBAR. Each value of P_{ij} is the probability of observing a frame of the λ state λ_i in the state λ_j . Probabilities off-diagonal indicate overlap between λ windows. It is desirable that the super and subdiagonals have all elements equal or greater than 0.03^3 (B) Convergence plot for forward and reverse trajectories of **FXR_81**→**FXR_85** using OPLS-AA/M/LigParGen and no REST2 in water. The gray area represents the standard deviation of the $\Delta\Delta G$ estimate using the entire trajectory. Stable predictions at longer simulation times suggest good convergence (i.e. that extending

the simulation would not change the results). (C) Empirical transition matrix estimated from observed replica exchanges. Each off diagonal element P_{ij} is the estimated probability of a swap between replicas i and j . Higher values in the superdiagonal and subdiagonal indicate increased transitions between replicas, which can lead to enhanced state mixing. The plot depicted in panel (C) is only available when HREX is used. (D) Hamiltonian occupancy for each replica. Each replica is represented by a different color, and the bar height is the probability that a given Hamiltonian (denoted by the x-axis index) will be found occupied by a given replica at any moment. More diversely colored bars indicate that Hamiltonians were sampled by more replicas. More transitioning between replicas can lead to improved state mixing, and, ideally, all replicas should explore all Hamiltonians^{4,5}. The plot depicted in panel (D) is only available when HREX is used.

MD systems using a 1.5 nm minimum distance to the box faces

For some perturbations (**Table S1**), the water leg systems were rebuilt using a minimum distance of 1.5 instead of 1.0 nm to the box faces.

Table S1 – Perturbations for which a minimum distance of 1.5 nm to the box faces was used.

Force field	REST2	Perturbation	Replicas
CHARMM36m/ CGenFF	No	FXR_10→FXR_79	2
		FXR_12→FXR_76	1
		FXR_12→FXR_77	3
		FXR_76→FXR_84	1
		FXR_76→FXR_89	1, 2
		FXR_81→FXR_82	2
		FXR_81→FXR_85	2
		FXR_82→FXR_84	1, 3
	Yes	FXR_76→FXR_84	1
		FXR_76→FXR_89	2, 3
		FXR_78→FXR_83	2
		FXR_81→FXR_85	2
CHARMM36-YF/ CGenFF	No	FXR_76→FXR_84	3
		FXR_76→FXR_89	2
		FXR_81→FXR_82	2
		FXR_81→FXR_85	1
		FXR_82→FXR_84	2
CHARMM36-WYF/ CGenFF	No	FXR_76→FXR_89	2
		FXR_82→FXR_84	3
		FXR_85→FXR_88	2

Table S2 – 12 λ scheme^a used to represent the perturbations in our study.

$\lambda =$	0	1	2	3	4	5	6	7	8	9	10	11
vdW A	1.00 000	1.00 000	1.00 000	1.00 000	1.00 000	0.67 479	0.45 630	0.32 525	0.24 741	0.18 978	0.11 850	0.00 000
vdW B	0.00 000	0.11 850	0.18 978	0.24 741	0.32 525	0.45 630	0.67 479	1.00 000	1.00 000	1.00 000	1.00 000	1.00 000
Coulomb A	1.00 000	0.75 000	0.50 000	0.25 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000
Coulomb B	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.25 000	0.50 000	0.75 000	1.00 000

^a The 12 λ windows for van der Waals and Coulomb interactions were the same as in FEP+.

Table S3 – 24 λ scheme used to represent the perturbations in our study.

$\lambda =$	0	1	2	3	4	5	6	7	8	9	10	11
vdW A	1.00 000	1.00 000	1.00 000	1.00 000	1.00 000	0.67 479	0.45 630	0.32 525	0.24 741	0.18 978	0.11 850	0.00 000
vdW B	0.00 000	0.11 850	0.18 978	0.24 741	0.32 525	0.45 630	0.67 479	1.00 000	1.00 000	1.00 000	1.00 000	1.00 000
Coulomb A	1.00 000	0.75 000	0.50 000	0.25 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000
Coulomb B	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.25 000	0.50 000	0.75 000	1.00 000
$\lambda =$	12	13	14	15	16	17	18	19	20	21	22	23
vdW A	1.00 000	1.00 000	1.00 000	1.00 000	1.00 000	0.67 479	0.45 630	0.32 525	0.24 741	0.18 978	0.11 850	0.00 000
vdW B	0.00 000	0.11 850	0.18 978	0.24 741	0.32 525	0.45 630	0.67 479	1.00 000	1.00 000	1.00 000	1.00 000	1.00 000
Coulomb A	1.00 000	0.75 000	0.50 000	0.25 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000
Coulomb B	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.00 000	0.25 000	0.50 000	0.75 000	1.00 000

Table S4 - Data for original submissions to D3R grand challenge 2 for the spiro series listed as using a full free energy method. Adapted from D3R website.

Submission ID	PI name / Submitter Name	Number of ligands	Kendall's Tau	RMSEc (kcal/mol)	Protocol Name from Protocol File	List of Software Packages Used
xk67c	Steve Jones / Oleksandr Yakovenko	18	0.62 ± 0.12	0.94 ± 0.12	Jarzinsky umbrella sampling, fset0	Jarzinsky umbrella sampling
81n55		18	0.55 ± 0.12	1.31 ± 0.20	free energy perturbation with Schrodinger fep/rest	Schrodinger-FEP
jzrt5	Julien Michel / Julien Michel	18	0.37 ± 0.14	1.45 ± 0.23	alchemical free energy protocol with average network analysis and mixed scaling of partial charges	FESetup, NetworkX
adxx8	Ulf Ryde, Alfonso Garcia-Sosa / Martin Olsson	18	0.48 ± 0.13	1.46 ± 0.20	amber pmemd free energy perturbation with charge correction	Amber
x2j7p	Zoe Cournia / Christina Athanasiou	18	0.50 ± 0.10	1.49 ± 0.19	Free energy perturbation calculations	Schrodinger-FEP

Submission ID	PI name / Submitter Name	Number of ligands	Kendall's Tau	RMSEc (kcal/mol)	Protocol Name from Protocol File	List of Software Packages Used
izhdb	Charles L. Brooks III / Xinqiang Ding	18	0.28 ± 0.17	1.61 ± 0.28	multi site lambda dynamics	CHARMM
qt771	Julien Michel / Julien Michel	18	-0.14 ± 0.20	4.11 ± 0.74	alchemical free energy protocol with average network analysis	FESetup, NetworkX
2ytv8	Bogdan Iorga / Bogdan Iorga	18	-0.15 ± 0.23	4.37 ± 1.35	Alchemical Free Energy Calculation	GROMACS

Table S5 - Statistics on individual FEP replicas using Amber03/GAFF, CHARMM36m/CGenFF, CHARMM36-YF/CGenFF, CHARMM36-WYF/CGenFF and OPLS-AA/M/LigParGen with and without REST2.

Force field	REST2	Replica	Kendall's τ^b	Spearman's r^b	Pearson's r^b	RMSE _c (kcal/mol)	Mean error ^c $\pm$ SD (kcal/mol)	MAE ^d $\pm$ SD (kcal/mol)
Amber03/ GAFF	No	1	0.52 $\pm$ 0.14	0.70 $\pm$ 0.16	0.67 $\pm$ 0.13	1.15 $\pm$ 0.16	0.20 $\pm$ 1.20	1.05 $\pm$ 0.61
		2	0.56 $\pm$ 0.12	0.74 $\pm$ 0.13	0.69 $\pm$ 0.11	1.17 $\pm$ 0.19	-0.18 $\pm$ 1.21	1.04 $\pm$ 0.64
		3	0.34 $\pm$ 0.16	0.56 $\pm$ 0.20	0.61 $\pm$ 0.16	1.16 $\pm$ 0.22	0.62 $\pm$ 1.19	1.11 $\pm$ 0.75
		Mean ^a	0.47 $\pm$ 0.12	0.67 $\pm$ 0.10	0.66 $\pm$ 0.05	1.16 $\pm$ 0.01	0.21 $\pm$ 1.24	1.06 $\pm$ 0.67
	Yes	1	0.30 $\pm$ 0.18	0.50 $\pm$ 0.23	0.53 $\pm$ 0.17	1.52 $\pm$ 0.25	1.03 $\pm$ 1.55	1.49 $\pm$ 1.12
		2	0.47 $\pm$ 0.14	0.66 $\pm$ 0.16	0.60 $\pm$ 0.12	1.54 $\pm$ 0.36	0.45 $\pm$ 1.60	1.28 $\pm$ 1.06
		3	0.65 $\pm$ 0.12	0.82 $\pm$ 0.12	0.68 $\pm$ 0.10	1.29 $\pm$ 0.30	0.99 $\pm$ 1.31	1.43 $\pm$ 0.82
		Mean ^a	0.47 $\pm$ 0.18	0.66 $\pm$ 0.16	0.61 $\pm$ 0.08	1.45 $\pm$ 0.14	0.83 $\pm$ 1.52	1.40 $\pm$ 1.01
CHARMM36m /CGenFF	No	1	-0.01 $\pm$ 0.25	0.00 $\pm$ 0.32	-0.01 $\pm$ 0.30	1.58 $\pm$ 0.30	1.22 $\pm$ 1.60	1.77 $\pm$ 0.96
		2	0.10 $\pm$ 0.23	0.24 $\pm$ 0.31	0.23 $\pm$ 0.30	1.29 $\pm$ 0.25	0.56 $\pm$ 1.33	1.13 $\pm$ 0.90
		3	0.41 $\pm$ 0.18	0.60 $\pm$ 0.22	0.52 $\pm$ 0.18	1.07 $\pm$ 0.17	1.43 $\pm$ 1.04	1.52 $\pm$ 0.90
		Mean ^a	0.16 $\pm$ 0.22	0.28 $\pm$ 0.30	0.24 $\pm$ 0.26	1.31 $\pm$ 0.25	1.07 $\pm$ 1.39	1.47 $\pm$ 0.96
	Yes	1	0.36 $\pm$ 0.19	0.45 $\pm$ 0.24	0.38 $\pm$ 0.21	1.13 $\pm$ 0.23	0.30 $\pm$ 1.16	0.92 $\pm$ 0.77
		2	0.12 $\pm$ 0.23	0.20 $\pm$ 0.30	0.26 $\pm$ 0.29	1.20 $\pm$ 0.22	0.38 $\pm$ 1.24	1.00 $\pm$ 0.82
		3	0.32 $\pm$ 0.20	0.44 $\pm$ 0.25	0.38 $\pm$ 0.26	1.28 $\pm$ 0.22	0.57 $\pm$ 1.32	1.18 $\pm$ 0.82
		Mean ^a	0.27 $\pm$ 0.13	0.37 $\pm$ 0.14	0.34 $\pm$ 0.07	1.20 $\pm$ 0.08	0.41 $\pm$ 1.25	1.03 $\pm$ 0.81
CHARMM36-YF /CGenFF	No	1	0.14 $\pm$ 0.22	0.26 $\pm$ 0.29	0.26 $\pm$ 0.29	1.37 $\pm$ 0.34	0.18 $\pm$ 1.42	1.06 $\pm$ 0.96
		2	0.10 $\pm$ 0.19	0.23 $\pm$ 0.26	-0.02 $\pm$ 0.26	1.53 $\pm$ 0.34	0.52 $\pm$ 1.62	1.39 $\pm$ 0.98
		3	-0.12 $\pm$ 0.23	-0.16 $\pm$ 0.31	-0.15 $\pm$ 0.3	1.57 $\pm$ 0.32	-0.10 $\pm$ 1.59	1.27 $\pm$ 0.95
		Mean ^a	0.04 $\pm$ 0.14	0.11 $\pm$ 0.24	0.03 $\pm$ 0.21	1.49 $\pm$ 0.11	0.20 $\pm$ 1.56	1.24 $\pm$ 0.97

Force field	REST2	Replica	Kendall's τ^b	Spearman's r^b	Pearson's r^b	RMSE _c (kcal/mol)	Mean error ^c $\pm$ SD (kcal/mol)	MAE ^d $\pm$ SD (kcal/mol)
CHARMM36-WYFF/CGenFF	No	1	-0.05 $\pm$ 0.23	-0.01 $\pm$ 0.31	-0.28 $\pm$ 0.31	1.97 $\pm$ 0.49	-0.11 $\pm$ 1.85	1.31 $\pm$ 1.31
		2	-0.19 $\pm$ 0.25	-0.17 $\pm$ 0.32	-0.34 $\pm$ 0.33	1.90 $\pm$ 0.52	0.02 $\pm$ 1.84	1.21 $\pm$ 1.39
		3	-0.05 $\pm$ 0.25	0.06 $\pm$ 0.33	-0.08 $\pm$ 0.29	1.82 $\pm$ 0.34	0.54 $\pm$ 1.56	1.39 $\pm$ 0.88
		Mean ^a	-0.10 $\pm$ 0.08	-0.04 $\pm$ 0.12	-0.23 $\pm$ 0.13	1.90 $\pm$ 0.07	0.15 $\pm$ 1.78	1.30 $\pm$ 1.22
OPLS-AA/M LigParGen	No	1	0.41 $\pm$ 0.19	0.55 $\pm$ 0.23	0.50 $\pm$ 0.22	1.18 $\pm$ 0.23	0.50 $\pm$ 1.21	1.02 $\pm$ 0.83
		2	0.27 $\pm$ 0.24	0.47 $\pm$ 0.29	0.43 $\pm$ 0.28	1.21 $\pm$ 0.26	-0.22 $\pm$ 1.25	0.92 $\pm$ 0.88
		3	0.41 $\pm$ 0.16	0.59 $\pm$ 0.20	0.60 $\pm$ 0.18	0.96 $\pm$ 0.16	-0.72 $\pm$ 0.98	0.98 $\pm$ 0.72
		Mean ^a	0.36 $\pm$ 0.08	0.54 $\pm$ 0.06	0.51 $\pm$ 0.09	1.12 $\pm$ 0.14	-0.15 $\pm$ 1.26	0.97 $\pm$ 0.81
	Yes	1	0.65 $\pm$ 0.17	0.75 $\pm$ 0.19	0.71 $\pm$ 0.17	0.92 $\pm$ 0.20	0.50 $\pm$ 0.94	0.78 $\pm$ 0.73
		2	0.41 $\pm$ 0.17	0.57 $\pm$ 0.23	0.34 $\pm$ 0.18	3.21 $\pm$ 1.41	-1.15 $\pm$ 3.32	1.68 $\pm$ 3.08
		3	0.52 $\pm$ 0.16	0.70 $\pm$ 0.19	0.59 $\pm$ 0.15	0.99 $\pm$ 0.18	-0.50 $\pm$ 1.01	0.79 $\pm$ 0.81
		Mean ^a	0.52 $\pm$ 0.12	0.67 $\pm$ 0.09	0.55 $\pm$ 0.19	1.71 $\pm$ 1.30	-0.39 $\pm$ 2.18	1.08 $\pm$ 1.94

^a Reported values are average $\pm$ standard deviation for the three replicas.

^b Statistics computed using the D3R evaluator script.

^c Mean of $\Delta\Delta G_{\text{expt}} - \Delta\Delta G_{\text{calc}}$. SD calculated over the 13 predictions.

^d One-sample Student's t-test for mean $\neq$ 0.0.

^e Mean Absolute Error.

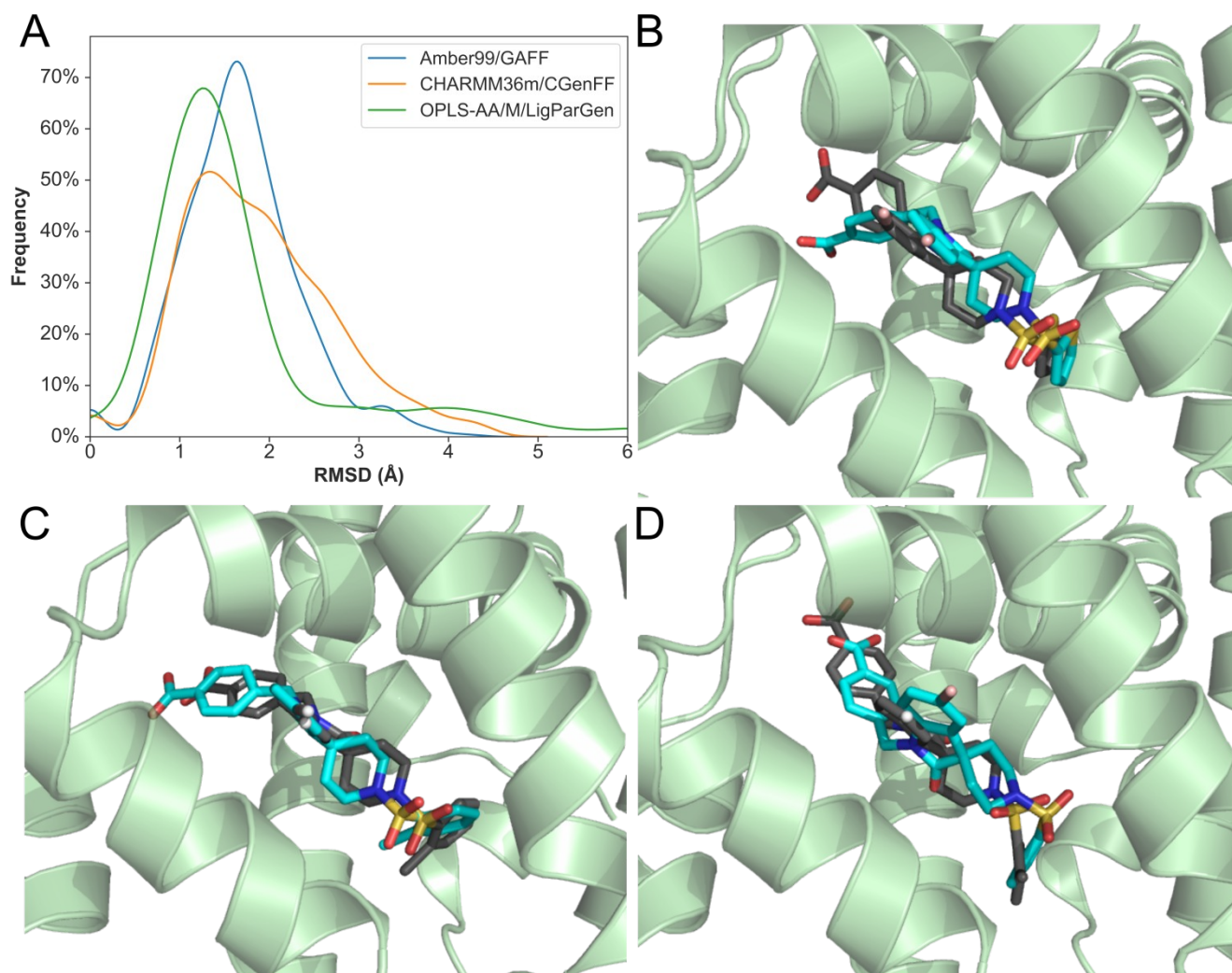


Figure S3 – Ligand RMSD and examples of some of the final ligand conformations. (A) Histogram of ligand RMSD observed for end-states during FEP MD after fitting using FXR C α positions. (B) Final frame of $\lambda=0$ for **FXR_10**→**FXR_76**, replica 1, CHARMM36m/CGenFF (RMSD = 2.16 Å). (C) Final frame of $\lambda=0$ for **FXR_12**→**FXR_77**, replica 1, CHARMM36m/CGenFF (RMSD = 1.60 Å). (D) Final frame of $\lambda=0$ for **FXR_85**→**FXR_88**, replica 3, CHARMM36m/CGenFF (RMSD = 1.99 Å). The crystal structure of FXR is represented in cartoon style (light green). The ligand pose at first MD frame is colored dark gray, and the ligand pose at the final MD frame is colored cyan. PyMOL was used for structure rendering.

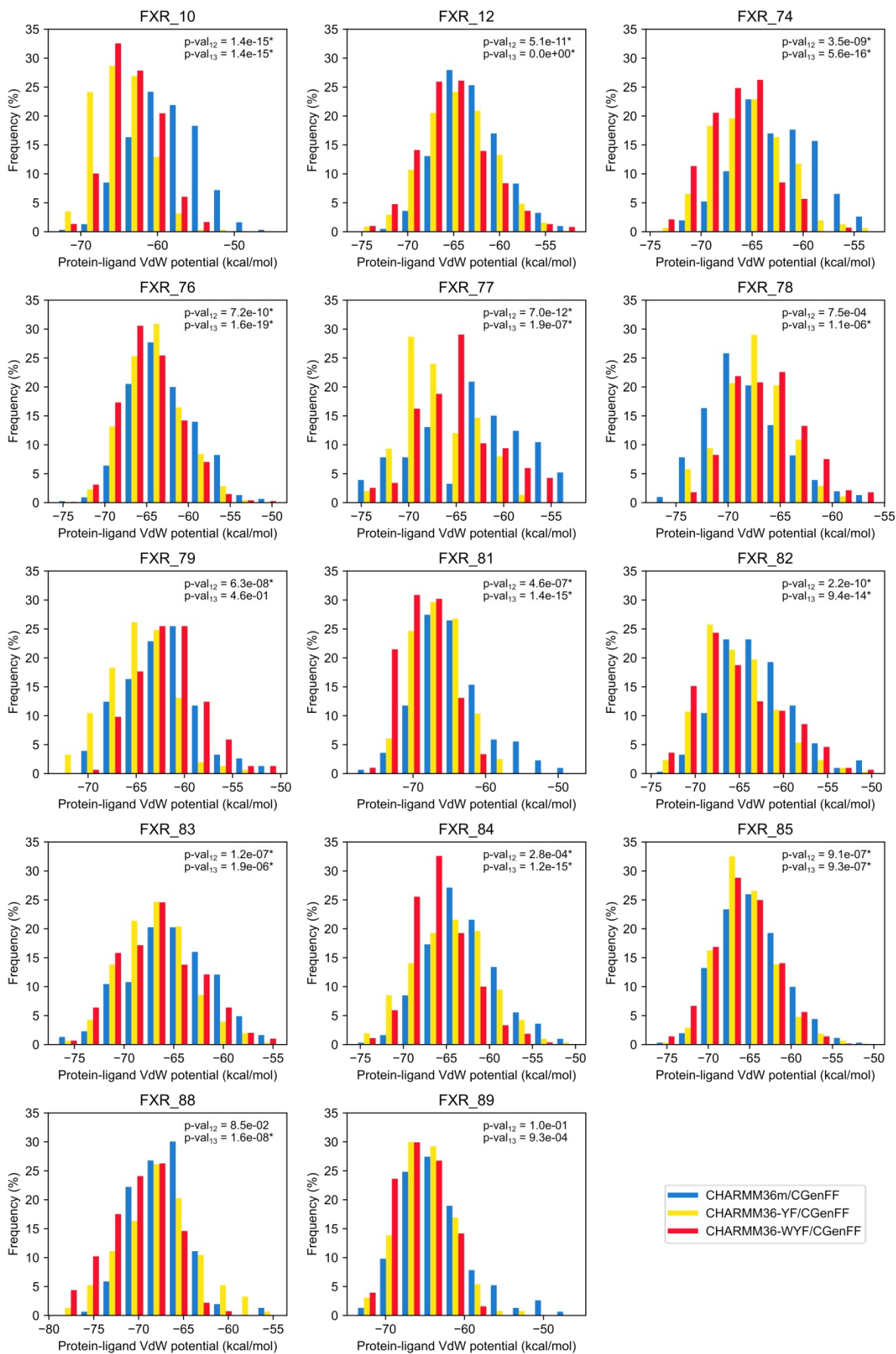


Figure S4 – Normalized histograms of van de Waals potentials between FXR and each of the spiro ligands extracted from the end-states ($\lambda = 0$ and $\lambda = 1$) of FEP simulations, obtained using GROMACS energy groups. $p\text{-val}_{12}$ is the two-sample Kolmogorov-Smirnov test p-value for the equality of CHARMM36m/CGenFF and CHARMM36-YF/CGenFF distributions. $p\text{-val}_{13}$ is the two-sample Kolmogorov-Smirnov test p-value for the equality of CHARMM36m/CGenFF and CHARMM36-WYF/CGenFF distributions. * denotes a statistically significant p-value ($\alpha = 0.01$), after a Bonferroni correction with $N=28$ was applied.

Table S6 – Cycle closure hysteresis observed using Amber03/GAFF, CHARMM36m/CGenFF, CHARMM-YF/CGenFF, CHARMM36-WYF/CGenFF and OPLS-AA/M/LigParGen, with and without REST2.

Force field	REST2	FXR_12-FXR_78-FXR-83		FXR_76-FXR_84-FXR_82-FXR_81-FXR_85	
		Absolute cycle closure error (kcal/mol)	Estimated standard deviation ^a (kcal/mol)	Absolute cycle closure error (kcal/mol)	Estimated standard deviation ^a (kcal/mol)
Amber03/GAFF	No	0.6	1.9	4.1	2.4
	Yes	1.1	2.5	4.5	3.2
CHARMM36m/ CGenFF	No	1.7	2.1	2.0	2.8
	Yes	2.0	1.8	3.4	2.4
CHARMM36-YF/ CGenFF	No	2.0	2.2	2.7	2.8
CHARMM36-WYF/ CGenFF	No	1.6	3.2	2.3	4.1
OPLS-AA/M/ LigParGen	No	1.9	1.5	0.2	2.0
	Yes	2.5	2.6	1.2	3.4

^a Estimated standard deviation of the cycle closure error, calculated as $\sqrt{N \sigma^2}$, where N is the number of perturbations in the cycle, and σ is the standard deviation of the errors (**Table 2**) (see Wang *et al.*⁶).

Comparing 12 and 24 λ windows for a problematic perturbation

The **FXR_10**→**FXR_79** perturbation had large standard deviations (2.7-16.5 kcal/mol), and errors (2.2-17.1 kcal/mol) with respect to the experimental data (see **Supporting Information 2** for data for each replica) when running 12 λ windows (**Table S2**). Using 12 λ and 20 ns sampling time yielded similar results as with 10 ns of sampling (**Table S7**).

Table S7 - Mean predicted $\Delta\Delta G$ with respect to **FXR_10**, standard deviations, and errors for **FXR_79** prediction obtained using 10 and 20 ns.

FEP sampling time (ns)	Amber03/ GAFF without REST2		CHARMM36m/ CGenFF without REST2		OPLS-AA/M/ LigParGen without REST2	
	Predicted $\Delta\Delta G \pm SD^a$ (kcal/mol)	Error to experimental value (kcal/mol)	Predicted $\Delta\Delta G \pm SD^a$ (kcal/mol)	Error to experimental value (kcal/mol)	Predicted $\Delta\Delta G \pm SD^a$ (kcal/mol)	Error to experimental value (kcal/mol)
10	3.4 $\pm$ 2.7	3.6	0.1 $\pm$ 16.5	0.3	2.0 $\pm$ 2.9	2.1
20	3.5 $\pm$ 1.7	3.7	6.8 $\pm$ 44.4	7.0	-2.5 $\pm$ 5.5	2.3

^a Mean predicted $\Delta\Delta G \pm$ standard deviation

The **FXR_10**→**FXR_79** transformation requires a change of the carboxylate position, and thus perturbation of charge localization (**Figure 3**). The charge hopping may increase the energy gap between windows, reducing overlap, and leading to convergence issues. The hypothesis was tested by expanding the number of λ windows used to represent the perturbation. **FXR_10**→**FXR_79** was rerun using 24 λ windows (**Table S3**) using all combinations of force fields, and in the presence and absence of REST2. A similar approach for perturbations involving charge-hopping was employed by Chen et al.⁷ when studying efficient protocols to allow charge perturbations and charge-hopping. Comparing the predictions using 24 and 12 λ windows shows that better agreement to experimental data was achieved with 24 λ windows. Both the mean absolute error (11.12 to 2.30 kcal/mol, Welch's t-test p-value = 0.0001 for equality of the means) and its standard deviation (8.87 to 2.37 kcal/mol Levene's test p-value

= 0.0002 for equality of the variances) decreased. This improvement suggests that a better sampling of the perturbation was achieved. Furthermore, the mean predicted $\Delta\Delta G$ changed from 1.91 ± 13.96 kcal/mol to -0.38 ± 3.51 kcal/mol using 12 and 24 λ windows, respectively. The change in the mean prediction was not statistically significant, but the decrease in the variance was (Welch's t-test p-value = 0.4528 for equality of the means; Levene's test p-value = 0.0002 for equality of the variances), suggesting that the predictions using 12 λ windows were not biased, but rather sampling from a population centered on the true value with a large variance. It is worth noting that a single prediction using 24 λ windows (OPLS-AA/M/LigParGen, and no REST2; see **Supporting Information 2**) had a large absolute error of 12.1 kcal/mol. Without this outlier, the mean absolute error of the **FXR_79** prediction using 24 λ windows would be 1.88 ± 1.17 kcal/mol (including the outlier, MAE = 2.30 ± 2.37 kcal/mol). To assess whether the source of the variation in the predictions was the poor overlap between adjacent windows, we analyzed the overlap matrix between states (**Figure S5a and b**). These matrices were calculated by PyAutoFEP using pymbar's implementation, and showed a better overlap between the states when using 24 λ windows. However, analyzing the same matrix for other perturbations, e.g. **FXR_10**→**FXR_76** (**Figure S5c**), it was not clear that **FXR_10**→**FXR_79** with 12 λ windows was substantially worse in this regard.

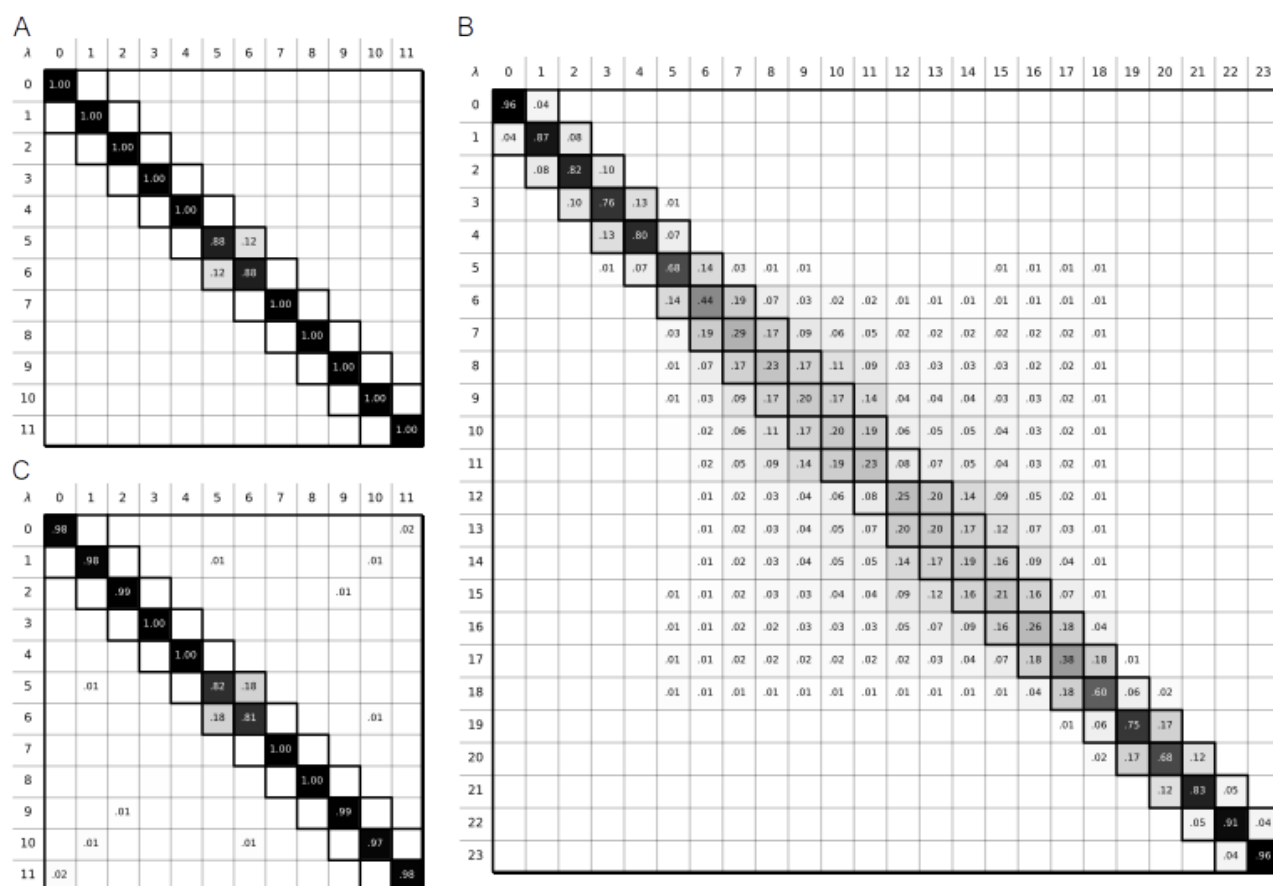


Figure S5 – Overlap matrices for the perturbation **FXR_10**→**FXR_79** using (A) 12 λ windows, and (B) 24 λ windows, and (C) for the perturbation **FXR_10**→**FXR_76** using 12 λ windows. All the matrices were obtained using Amber03/GAFF and no enhanced sampling.

The scaling values used for the 12 λ windows for van der Waals and Coulomb interactions were the same as in FEP⁸. The λ scheme is likely optimized for Schrödinger’s modified OPLS force field, and possibly not ideal for the different force fields employed in this work, leading to poor overlap between adjacent states. A test using 24 λ windows for all perturbations was performed using the Amber03/GAFF force field and no REST2, and similar results to the 12 λ calculations were obtained (Table S8).

Table S8 – Statistics on FEP replicas using Amber03/GAFF without REST2 using 12 or 24 λ windows.

Number of λ windows	Replica	Kendall's τ^b	Spearman's r^b	Pearson's r^b	RMSEC ^b (kcal/mol)	MAE ^c $\pm$ SD (kcal/mol)
12	1	0.52 ± 0.14	0.70 ± 0.16	0.67 ± 0.13	1.15 ± 0.16	1.05 ± 0.61
	2	0.56 ± 0.12	0.74 ± 0.13	0.69 ± 0.11	1.17 ± 0.19	1.04 ± 0.64
	3	0.34 ± 0.16	0.56 ± 0.20	0.61 ± 0.16	1.16 ± 0.22	1.11 ± 0.75
	Mean ^a	0.47 ± 0.12	0.67 ± 0.10	0.66 ± 0.05	1.16 ± 0.01	1.06 ± 0.67
24	1	0.45 ± 0.18	0.66 ± 0.20	0.66 ± 0.17	1.09 ± 0.20	0.94 ± 0.72

^a Reported values are average $\pm$ standard deviation for the three replicas.^b Statistics computed using the D3R evaluator script.^c Mean absolute error.

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